

Mohamed Aamir

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PROFESSIONAL SUMMARY

AI-focused Software Engineering undergraduate with strong foundations in Data Structures, Algorithms, and System Design. Ranked 164 out of 8365 registered university candidates on GeeksforGeeks (Top 2%). Experienced in building production-grade AI systems using LLMs, RAG pipelines, and scalable backend architectures. Skilled in designing end-to-end systems spanning data ingestion, retrieval, and low-latency APIs, with a focus on deploying AI solutions in real-world, distributed environments.

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, SQL
AI/ML: LLMs, RAG, NLP, Prompt Engineering, Transformers
Backend: FastAPI, Flask, REST APIs, System Design
Databases: Firebase, DynamoDB, Relational Databases
Cloud: AWS (Lambda, S3, DynamoDB, CloudFront), Serverless Architecture
Tools: Git, Docker, FAISS, Postman
Core CS: Data Structures, Algorithms, OOP, Operating Systems, DBMS

PROJECTS

- Multilingual RAG-based Government Scheme Assistant | *LLMs, FAISS, FastAPI*** 2026
- Designed a production-grade multilingual RAG pipeline for grounded, context-aware responses, reducing hallucinations.
 - Engineered a scalable ingestion and indexing pipeline for PDFs/web data, enabling efficient retrieval over large datasets.
 - Implemented FAISS-based vector search with optimized top-k retrieval, improving response relevance and accuracy.
 - Built a modular FastAPI backend for low-latency, concurrent retrieval and inference workflows.
 - Developed a multi-modal (text + voice) Telegram interface, improving accessibility for diverse users.
- AI-driven Elderly Care Platform | *React Native, FastAPI, Gemini*** 2025
- Designed a hybrid conversational AI system combining rule-based logic with Gemini LLM for reliable healthcare assistance.
 - Architected a modular FastAPI + Firebase backend with secure, role-based access control for real-time data synchronization.
 - Built RESTful APIs for real-time health data, chat, and notifications, enabling seamless caregiver-patient communication.
 - Designed event-driven alerting and monitoring pipelines, improving caregiver response time and coordination.
 - Built a scalable cross-platform mobile system with extensible architecture for future ML integrations.
- Scalable Serverless Music Streaming Platform | *AWS, Distributed Systems*** 2025
- Architected a serverless streaming system (AWS Lambda, S3, DynamoDB, CloudFront) for low-latency media delivery.
 - Designed distributed caching and indexing mechanisms, reducing streaming latency from ~ 500ms to <100ms.
 - Built scalable backend APIs and data models supporting 100K concurrent users with auto-scaling and fault tolerance.
 - Optimized authentication, metadata indexing, and query pipelines, improving content retrieval speed and throughput.
 - Designed an AI-driven recommendation system architecture to enable personalized user experiences.
- Skin Cancer Detection & Report Summarization | *CNNs, LLMs, Flask*** 2024
- Built an end-to-end AI diagnostic pipeline combining CNN-based classification and LLM-powered summarization.
 - Trained and optimized a CNN on HAM10000 dataset (~ 92% accuracy) for reliable skin disease detection.
 - Designed a multi-stage inference pipeline (image → diagnosis → summarization) generating structured medical insights.
 - Developed a Flask-based backend for real-time image processing and model inference.
 - Reduced diagnostic turnaround time by ~ 70% through automated AI workflows, improving efficiency of medical analysis.

EDUCATION

College of Engineering Guindy, Anna University	Chennai, India
<i>B.Tech in Information Technology</i>	2022 – 2026

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
Machine Learning for Data Science – IBM
AWS Academy Cloud Foundations – Amazon Web Services

ACHIEVEMENTS

Top 2% (164/8365) university rank on GeeksforGeeks coding platform
Selected for **buildspace** program (competitive builder community)
Participated in 4+ hackathons, building rapid prototypes under time constraints